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Reading Skills in Hyperlexia: A Developmental Perspective

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Hyperlexia is characterized by advanced word-recognition skills in individuals who otherwise have pronounced cognitive, social, and linguistic handicaps. Language, word recognition, and reading-comprehension skills are reviewed to clarify the nature and core deficits associated with the disorder. It is concluded that hyperlexia should be viewed as part of the normal variation in reading skills, which are themselves associated with individual differences in phonological, orthographic, and semantic processing, short-term memory, and print exposure. A compulsive preoccupation with reading may also be crucial to the development of a hyperlexic reading profile. A theoretical framework, based on recent connectionist models of reading development, is described. This perspective provides a satisfactory account for how individual differences in a number of different skills can lead to a variety of manifestations of reading behavior, including hyperlexia.

In the normal population, children's ability to decipher print generally develops alongside their ability to understand what they are reading. In cases of developmental disorder, however, the developmental course of different component reading skills may be out of step. For example, dyslexic children have normal-range intellectual abilities but have difficulty learning to decode printed words. In contrast to their weak word recognition, many dyslexic children comprehend well (e.g., Shaywitz, 1996; Snowling, 1987). The focus of this review is on the reading skills of a group of children who have comprehension difficulties yet possess well-developed word-recognition skills—so called hyperlexic children.[1] *Hyperlexia,* a term first used by Silberberg and Silberberg (1967), refers to a disorder characterized by the development of advanced word-recognition skills in individuals who otherwise have pronounced cognitive and language deficits. In addition to a discrepancy between word recognition and verbal comprehension/cognitive ability, studies have highlighted the early onset of word recognition in hyperlexia, often occurring without instruction and before any expressive language develops. Hyperlexia has also been associated with other developmental disorders, most notably autism-spectrum disorders.

Perhaps the central question of interest is how hyperlexic children develop exceptional word-recognition skills, with very little direct instruction and despite the presence of wide-ranging (and in some cases profound) cognitive, linguistic, and social handicaps. Two rather different perspectives on the question can be discerned from the literature. Some view hyperlexia as a primary disorder, a distinct syndrome consisting of three symptoms, namely, impaired comprehension, exceptional word recognition, and the spontaneous development of reading before 5 years of age (e.g., Aaron, 1989; Healy, 1982). Others (Graziani, Brodsky, Mason, & Zager, 1983; McClure & Hynd, 1983; Pennington, Johnson, & Welsh, 1987; Rispens & Van Berckelaer, 1989) use the term hyperlexia to refer to a particular reading profile in which word-recognition skills are substantially ahead of comprehension and other cognitive skills. On this view, hyperlexia is a behavioral characteristic, not a discrete diagnostic category.

As a consequence of the term hyperlexia being used in these two different ways, the existing literature is confusing. The aim of this review is to provide an integrated account of the existing literature within a framework taken from current connectionist models of learning to read. To fulfill this aim, I discuss and accommodate four sources of evidence:

1. The relationship between the underlying cognitive and linguistic skills that characterize hyperlexia and the reading behavior that emerges.
2. The relative integrity and deficiency of different reading component skills in hyperlexia.
3. The unusual preoccupation with reading that is often seen in hyperlexia.
4. The association between hyperlexia and other developmental disorders.

Before reviewing this evidence in detail, it is useful to begin by considering the nature of the task of learning to read. With this as a backdrop, it is possible to assess to what extent hyperlexic children achieve skilled reading and to consider how their accomplishments develop in the face of cognitive, linguistic, and social handicaps.

Learning to Read: A Task Analysis

Reading is a complex task: To read even a simple sentence demands a number of skills ranging from recognizing each indi-

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[1] For the purposes of this review, word recognition is defined as the ability to pronounce single words presented out of context; the ability to understand printed words and texts is referred to as reading comprehension.

vidual word to understanding the intended meaning of the text. Although these components are commonly described separately, it is important to remember that in skilled reading, they are highly interactive and proceed in a more or less parallel fashion (McClelland & Rumelhart, 1981; Stanovich, 1980).

Word Recognition

Essentially, word recognition involves learning associations between visual stimuli and linguistic information. Children need to learn not only the visual pattern associated with each individual letter (Gibson & Levin, 1975), they also need to develop abstract orthographic representations to recognize letters regardless of their precise visual appearance (e.g., lowercase or uppercase, typeface or handwritten). It is also possible that some letter patterns (e.g., *ght, th*) and very high frequency words (e.g., *the*) are processed in memory as orthographic units. Orthographic information is important from an early stage of reading development: Children who are sensitive to orthographic factors tend to be better readers than children who are less sensitive (Gottardo, Siegel, & Stanovich, 1998), and for children with a reading age beyond 7 years, reducing the quality and availability of orthographic information seriously disrupts text reading efficiency (Snowling & Frith, 1981).

In an alphabetic language such as English, orthographic units are associated with spoken sounds, and evidence suggests that children are also sensitive to the sound properties of written words from an early age. For example, 5-year-old beginner readers learn to associate the cue letter string *lft* with the spoken form "elephant" faster than a cue letter string consisting of visually distinctive letters such as *bxj* (Ehri & Wilce, 1985; Rack, Hulme, Snowling, & Wightman, 1994).

The alphabetic nature of English orthography poses two problems to the beginning reader. First, phonemes are abstract properties of speech (Liberman, Cooper, Shankweiler, & Studdert-Kennedy, 1967) that young children are not usually aware of (Liberman, Shankweiler, Fischer, & Carter, 1974). To understand that the letter *P* represents the abstract phoneme /p/, children must be sympathetic to the fact that *piece, pot,* and *pit* all contain /p/, even though acoustically they may sound slightly different. This skill has been termed *phonological awareness* and is considered to be crucial for the development of skilled and automatic word recognition (Brady & Shankweiler, 1991; Goswami & Bryant, 1990). A vast amount of evidence shows that preschool phonological skills are strong predictors of later reading achievement (Bradley & Bryant, 1983; Stanovich, Cunningham, & Cramer, 1984), and as a corollary, dyslexic children have well-documented difficulties across a range of phonological-processing tasks (Snowling, 1991).

In summary, certain levels of visual/orthographic skills and phonological skills are considered necessary for skilled reading to develop. Studies to be reviewed later have questioned whether exceptional strengths or weaknesses in these basic skills can account for the pattern of reading behavior seen in hyperlexia.

Even if children are able to recognize letters and "hear out" sounds in words, they are still faced with the problem that, in English, there is not a one-to-one relationship between letters and sounds. Although English is alphabetic, it is not transparent; some letter patterns are pronounced differently in different words (e.g., *cough, tough, bough*), and even letter patterns that appear regular can be inconsistent (e.g., *hint, lint, mint* vs. *pint*). Some theorists have argued that this feature of the orthography demands that two different systems are used in reading. According to such a dual-route view (e.g., Coltheart, Patterson, & Marshall, 1980), there are two main routes to word recognition. In the indirect *phonological* route, words are read using grapheme–phoneme correspondence rules, thus enabling regular words and nonwords to be pronounced correctly. Alongside this route, the direct *lexical* route activates word-specific representations, allowing exception words to be read correctly. However, as the direct route consists of word-specific information, it provides no facility for reading nonwords. Potentially, hyperlexia may be the result of over-reliance on either of these routines; evidence for this possibility is reviewed later.

An alternative to the dual-route view is that all words are recognized by the same system. This view is supported by more recent connectionist accounts of reading development (Plaut, McClelland, Seidenberg, & Patterson, 1996; Seidenberg & McClelland, 1989). These models consist of many simple processing input and output units that are linked together via weighted connections. As a function of learning, the weights on the connections gradually change so that the architecture comes to embody what is known about the environment, and this knowledge is distributed throughout the network. In short, the network learns the statistical regularities inherent in the learning environment, and as a consequence, it is able to generalize from its training vocabulary to read novel and unfamiliar words. As English is a quasi-regular language (it is essentially regular but it permits some exceptions), this is reflected in the model's knowledge: It exhibits both rule-like and word-specific behavior without two separate sets of processes being specified.

Regardless of whether one adopts a dual-route or a connectionist framework, children need to learn to make associations between print and sound. Given adequate visual, orthographic, and phonological-processing skills, other factors may also play a role in the development of skilled word recognition. Long-term memory is needed for rapid and automatic retrieval of stored information, and verbal short-term memory is needed to allow children to blend and segment words as they read and spell (Wagner & Torgeson, 1987).

Another factor contributing to the development of fast and efficient word recognition is adequate practice. Measures of *print exposure,* a term used to index a child's familiarity and experience of written words, account for significant variation in word recognition, even after variation in phonological skill is partialled out of the regression equation (e.g., Cunningham & Stanovich, 1991). In short, children who read more are better readers than children who read less. This observation fits well with connectionist models of reading in which learning is shaped by environmental experience, that is, the range of words encountered by the network and the number of times each has been encountered.

Reading Comprehension

Successful reading comprehension depends on a number of processes that can be broadly separated into two classes: local and text-level processing. At a local level, the meanings of individual words need to be understood; texts that contain difficult vocabulary are difficult for children to understand (Beck, Perfetti, & McKeown, 1982), and children who have poor vocabulary skills are also poor at comprehending text (Nation & Snowling, 1998a). Moreover, as well as activating contextually relevant information, children need to learn to suppress irrelevant information. Gernsbacher and Faust (1991) presented students with a sentence context to read and then asked them to decide whether a target word fit the meaning of the preceding sentence. When the target word was presented immediately after the sentence, all participants were slower to reject ACE following "He dug with the *spade*" compared with when ACE followed "He dug with the *shovel*." This suggests that the contextually irrelevant meaning of spade was activated, despite the biasing context. However, when there was an 850-ms delay between reading the sentence and seeing the target word, skilled readers no longer showed interference. The less-skilled readers, however, maintained heightened activation for the contextually inappropriate interpretation. Thus, skilled readers very quickly suppress irrelevant information and focus attention on contextually important information. In contrast, less-skilled readers are less adept at suppressing contextually inappropriate meanings of words.

Understanding the contextually relevant meaning of each individual word, however, is no guarantee that a text will be accurately comprehended. To ensure that comprehension flows from one phrase to the next, text-level processes are needed. In the sentence "John gave the toy to Tom because it was his birthday," readers need to integrate successive propositions to understand that "his" probably refers to Tom, not John. Adequate comprehension also requires the reader to go beyond what is explicitly stated; relevant pieces of knowledge need to be linked and inferences made to fill in information that is only implicit in the text, for example, that people give presents on birthdays, or that as Tom was given a toy, he is likely to be a child. In short, comprehension is a constructive, integrative process in which skilled readers build an active and dynamic mental representation of the text as they read.

Children may fail to build such a mental model for a number of reasons. As discussed above, they may simply fail to understand the vocabulary, or they may not possess the relevant schematic knowledge (Perfetti, 1985). In addition, some children have difficulties engaging in constructive processing and making inferences, despite possessing the relevant knowledge (Yuill & Oakhill, 1991). Finally, many of the processes involved in comprehension are likely to be influenced by memory: Words and propositions need to be remembered while semantic and syntactic information is activated and an integrated and coherent representation of the text is constructed. Thus, the limits of verbal memory may place constraints on comprehension (Just & Carpenter, 1992).

A number of metacognitive processes are also considered to be important for skilled reading comprehension (Baker & Brown, 1984). Very broadly, the term *metacognition* refers to a reader's own awareness and strategic control over the reading process, including the ability to allocate attention to relevant information and to identify important aspects of text. One important metacognitive skill is comprehension monitoring, which refers to the ability to monitor one's own comprehension to assess whether understanding is satisfactory. Young children are poor at comprehension monitoring, as are children with reading comprehension difficulties (Baker & Brown, 1984; Paris & Myers, 1981; Yuill & Oakhill, 1991). Thus, these children are less likely to notice their failure to comprehend, and as a result, they are less likely to take corrective "fix-up" action such as rereading the text.

In summary, skilled reading demands a variety of perceptual, linguistic, and cognitive skills. Although not captured by cognitive models of the reading process, other factors such as motivation, attitude to reading, amount of practice, and relevant opportunities for learning are also likely to shape an individual's reading development.

Reading Skills in Hyperlexia

The key feature of hyperlexia is word-recognition skills that are well in advance of comprehension. Hyperlexic reading is also remarkable because of its early onset (reading can begin as early as 2 or 3 years of age, without any formal instruction) and the compulsive preoccupation that many hyperlexic children have with reading (Aram & Healy, 1988). Hyperlexic children tend to read more rapidly than controls, and moreover, scrambling word order in text disrupts their reading speed far less relative to normal readers (O'Connor & Hermelin, 1994).

Perhaps the first question to be asked is how remarkable are the word-recognition skills of hyperlexic children? Surprisingly, no operational definitions exist, although in Richman and Kitchell's (1981) study, only average IQ hyperlexic children who were reading at least 2 years above their present grade level were included. Welsh, Pennington, and Rogers (1987) used a Reading Quotient (RQ) index (reading age divided by mental age) to provide an objective measure of word-recognition precocity, relative to overall general intellectual ability. Thus, an RQ of 1 indicates word-recognition skills that are consistent with IQ, whereas an RQ of less (or more) than 1 indicates reading skills that are weaker (or greater) than predicted by IQ. Because an RQ of less than 0.80 is usually taken as indicative of specific reading disorder or dyslexia, an RQ greater than 1.20 would be indicative of hyperlexia. Indeed, all 5 participants in their study obtained an RQ considerably in advance of 1.20 (range 1.37 to 2.80). Welsh et al. (1987) also calculated RQs for other hyperlexic children on the basis of data reported in other studies (Healy, Aram, Horwitz, & Kessler, 1982; Richman & Kitchell, 1981; Siegel, 1984) and found that virtually all of the cases had RQs well in advance of 1.20. Finally, although Welsh et al. calculated RQ relative to mental age, it is worth noting that in their study, all of the children had word-recognition skills at least consistent with chronological age and, indeed, 4 out of the 5 children considerably in advance of the levels expected from chronological age.

Thus, when objective criteria such as RQ indexes are used, hyperlexic children do demonstrate exceptional word-recognition skills. Clearly, the next question to address is how hyperlexic children achieve such surprisingly good word-recognition skills. An appropriate starting point is to examine the underlying cognitive and linguistic abilities typically associated with hyperlexia. If hyperlexic children have particular deficits in certain cognitive or linguistic processes, coupled with particular strengths in other areas, this may provide some explanation for the surprising pattern of their reading behavior.

General Cognitive and Linguistic Skills in Hyperlexic Children

The majority of experimental and clinical reports point to at least some degree of mental retardation in most hyperlexic children. A number of case studies describe individuals with full-scale IQ scores falling below 60 (Aram, Rose, & Horwitz, 1984; Glosser, Friedman, & Roeltgen, 1996; Siegel, 1984). However, there is considerable variation in the literature, and this is well-illustrated in group experimental studies (Cohen, Hall, & Riccio, 1997; Fontenelle & Alarcon, 1982; Goldberg & Rothermel, 1984; Healy et al., 1982; Huttenlocher & Huttenlocher, 1973; Richman & Kitchell, 1981; Silberberg & Silberberg, 1971). For example, Healy et al. (1982) reported a full-scale IQ range of 62–91, whereas Richman and Kitchell (1981) described 10 high-functioning hyperlexic children all with average general ability (IQ range 92–116).

Hyperlexic children have generally been reported to achieve significantly higher Performance scores than Verbal scores, and in some cases, the difference between Performance and Verbal IQ is substantial. In Richman and Kitchell's (1981) study, 9 of the 10 children showed a discrepancy of more than one standard deviation between Verbal and Performance IQ. However, some hyperlexic children do not show a Performance IQ advantage (Fontenelle & Alarcon, 1982; Welsh et al., 1987), and moreover, the divide between Performance and Verbal ability appears too gross to fully capture the cognitive profile typically found in hyperlexia. Examining performance across the subtests of the IQ battery, researchers have found that hyperlexics tend to perform best on those subtests requiring rote memory (Digit Span) or perceptual skills (Block Design). In contrast, performance is lowest on subtests requiring decision making, categorization, reasoning, or judgment (e.g., Comprehension, Picture Completion, Picture Arrangement). Welsh et al. (1987) calculated a verbal-comprehension factor, derived from four Weschler Intelligence Scale for Children IQ subtests (Comprehension, Similarities, Vocabulary, and Information) for 5 hyperlexic children. All of the children performed below average on this factor, indicating weaknesses in the verbal domain. Interestingly, this pattern of strengths and weaknesses is consistent for hyperlexic children of very low ability (Cobrinik, 1974) and also for those children of average ability (Richman & Kitchell, 1981).

Consistent with this general cognitive profile of weaknesses in the verbal domain, all of the reports that provide information on the language and communication skills of hyperlexic children detail substantial deficits in these areas. As Aram and Healy (1988) have noted, most hyperlexics initially come to the attention of professionals as a result of marked language abnormalities; many have a diagnosis of language impairment or a diagnosis that encompasses language delay such as autism-spectrum disorders. Virtually all studies document delay in the onset of using single words and connected speech, severe comprehension deficits as well as disordered expressive language, including echolalia (Cobrinik, 1974; Goldberg & Rothermel, 1984; Healy et al., 1982; Huttenlocher & Huttenlocher, 1973; Mehegan & Dreifuss, 1972). Elliot and Needleman (1976), among others, have described hyperlexic children who learned to read words before any spontaneous speech developed. Prosodic abnormalities have been noted in those hyperlexic children who do develop connected speech, including unusual intonation patterns and an abnormal rate of speech (Goldberg & Rothermel, 1984; Healy et al., 1982). Few hyperlexic children ever produce spontaneous speech or initiate conversation, and for those children that do, inappropriate interpersonal communication at a pragmatic level is common (Aram et al., 1984; Mehegan & Dreifuss, 1972).

Despite their language and communication problems, it is not clear to what extent hyperlexic children have speech difficulties. Aram and Healy (1988) suggested that although early speech development may be impaired, speech abnormalities are not a fundamental characteristic of hyperlexia. The fact that many hyperlexic children show echolalia is also consistent with the view that their speech skills are relatively intact. Interestingly, echolalia is more common in autistic children with hyperlexia than in autistic children who are not hyperlexic (Tirosh & Canby, 1993). Thus, it may be the case that many children with hyperlexia have satisfactory speech skills (at the level of phonology), despite pervasive difficulties with language and communication.

As described earlier in the task analysis of reading, successful reading comprehension requires knowledge of word meanings and schema as well as various integrative, inference making, and monitoring skills. Given the profile of severe delays and disorders in spoken language, it is likely that hyperlexic children will have reading-comprehension difficulties. Alongside these weaknesses, however, relative strengths in rote memory, perceptual processing, and perhaps even phonological (speech-based) processing may provide a reasonable substrate for learning print-pronunciation links. To examine this possibility, we review hyperlexic children's skills in two major components of word recognition, namely, visual/orthographic analysis and phonological processing.

Visual Analysis and Pattern Recognition

It is tempting to suggest that as a consequence of strong visual-spatial and visual associative memory skills, hyperlexic children read words as unanalyzed whole visual patterns. Certainly, the notion that hyperlexics recognize words by responding to the overall shape or gestalt of the word is consistent with their relative strengths in Block Design, as well as their demonstrably good memory for unrelated auditory and visual stimuli (Cain, 1969; Goodman, 1972). Cohen et al. (1997) reported an interesting comparison between children with specific language impairment (the SLI group) and language-impaired children who were also hyperlexic (the SLI + H group). The two groups did not differ on measures of expressive and receptive language or auditory verbal memory. However, the SLI + H group performed better than the SLI group on standardized tests of visuospatial skills and visual memory (as assessed by the Kaufman Assessment Battery for Children, Kaufman & Kaufman, 1983, and the Test of Visual-Perceptual Skills, Gardner, 1985). Cohen et al. suggested that these strengths may explain their elevated word-recognition skills.

However, a number of lines of evidence suggest that hyperlexic children do not rely solely on a process of simple pattern recognition when reading words. Goldberg and Rothermel (1984) presented hyperlexic children with visually distorted words to read (e.g., words presented in alternating case, vertical orientation, or with plus signs between letters). They reasoned that if hyperlexic children read by recognizing visual patterns, distorting the word forms in this manner should impair reading. Reading accuracy level was reduced only in the plus-sign condition, leading Goldberg and Rothermel to argue that hyperlexic reading was not accomplished by gestalt pattern recognition.

However, these results are difficult to interpret as it is not clear whether the children could read all of the target words when presented in normal script, and more important, no control children were tested. Plausibly, hyperlexic children's reading may be influenced more by visual disruptions than control children of equivalent reading age; this possibility remains to be tested in a satisfactory manner. Despite this limitation, Goldberg and Rothermel's finding that hyperlexic children are able to recognize letters despite changes in case or orientation shows that they have developed abstract orthographic representations: It is not the case that they read by a simple over-reliance on rote memory for visual configurations.

Perhaps the strongest evidence that hyperlexic children do not read by recognizing words holistically is provided by the observation that most hyperlexic children are able to read novel words accurately (e.g., Snowling & Frith, 1986). Thus, although hyperlexic children tend to have relative strengths in some aspects of visual processing (which they may well capitalize on for word recognition), their reading is not accomplished solely by visual pattern recognition.

Phonological-Processing Skills and Phonological Awareness

As discussed in the task analysis, phonological skills play a crucial role in the development of word recognition (Brady & Shankweiler, 1991; Goswami & Bryant, 1990). Given hyperlexic children's advanced word-recognition skills and the self-taught nature of their abilities, it seems reasonable to predict them to have satisfactory underlying phonological skills. Before discussing the results of studies investigating phonological skills in hyperlexia, it is important to note that in many studies of normal development, phonological skills have usually been measured by tests of phonological awareness (the ability to reflect on and manipulate the sound structure of spoken words). However, phonological awareness is only an indirect measure of the underlying phonological-processing skills that are important for literacy acquisition (Snowling & Hulme, 1994). As a consequence of low verbal ability, it is possible that hyperlexic children may perform poorly on those phonological awareness tests that require a higher level of metalinguistic ability, even if their underlying phonological skills are sound. A related issue is that a number of different measures of phonological awareness have been developed, all with varying cognitive and linguistic demands. Some of these measures are thought to tap skills that are prerequisites to learning to read, whereas others are thought to be more a consequence of having learned to read an alphabetic script (Perfetti, Beck, Bell, & Hughes, 1987). Thus, the literature documenting the relationship between phonological skills and learning to read in normal development presents a confusing background from which to assess the skills of this clinical population.

Goldberg and Rothermel (1984) measured phonological awareness in hyperlexic children by asking them to read pairs of words aloud and then to judge whether underlined segments sounded the same or different (e.g., *ch*ild and *ch*oice; *c*all and *s*ay). Some of the segments were phonologically identical but orthographically dissimilar (e.g., cr*y* and m*i*le), whereas others were orthographically identical but phonologically dissimilar (e.g., *g*oal and *g*em). Of the 8 children tested by Goldberg and Rothermel, only 3 were able to understand the requirements of the task, and even for these children, there was a bias toward responding "different" to all items. This result suggests that their phonological skills were poor. However, as with all of the tasks used by Goldberg and Rothermel, these results are difficult to interpret because of the lack of control data from normally developing children. Moreover, their test of phonological processing is confounded by the use of orthography and is made particularly difficult (and potentially confusing) by the inclusion of mismatches between orthography and phonology, a factor known to influence the ease of phonological processing in normal children and adults (Rack, 1985; Seidenberg & Tannenhaus, 1979).

Sparks (1995) asked 3 hyperlexic children to complete a battery of phonological tests, including the Lindamood Auditory Conceptualization Test (Lindamood & Lindamood, 1979) and 10 different phonological measures originally used by Stanovich, Cunningham, and Cramer (1984). These materials are excellent predictors of first-grade reading skills (Stanovich et al., 1984), and they include a variety of different tasks tapping different levels of phonological awareness. In the Lindamood test, children have to perceive and discriminate isolated sounds in a spoken sequence, and as with Goldberg and Rothermel's (1984) task, same–different judgments have to be made. All 3 hyperlexic children performed well below age-expected levels on the Lindamood tests.

These results are difficult to interpret. Perhaps the first point to note is the considerable range and spread of scores, both within and between each child. No consistent pattern of performance emerged, and each of the 3 children showed a different profile of strengths and weaknesses.

Interestingly, all 3 hyperlexics performed at ceiling when deleting an initial consonant, whereas this task was most difficult for the first-grade children in Stanovich et al.'s (1984) study. In contrast, 2 of the hyperlexic children found the rhyme tasks problematic, yet these were easiest for normally developing children. These observations are consistent with Seymour and Evans' (1992) case study of MP, a 6-year-old hyperlexic boy. Although MP was unable to perform rhyme-level tests, he performed well at phoneme deletion, a task that is considered to be among the most difficult of phonological awareness tests (Goswami & Bryant, 1990; Stanovich et al., 1984). However, performance on phoneme-level tasks is itself a product of acquiring a reasonable level of literacy (Perfetti et al., 1987). Hence, the difficulty of initial phoneme deletion for normally developing first graders can be understood, and plausibly, hyperlexic children's good performance on this task may be a by-product of their advanced word-recognition skills. By choosing experimental stimuli that contain a mismatch between the number of phonemes and the number of letters (e.g., *ship* contains four letters but only three phonemes), it would be possible to check to what extent hyperlexic children rely on alphabetic knowledge in phoneme-awareness tasks. In this example, a child relying on orthographic knowledge to delete the first phoneme of *ship* would produce the error "hip" rather than the correct answer "ip."

Less clear is why some hyperlexic children have difficulty with rhyme. One possibility is that rhyme-level tasks are more conceptually demanding than phoneme-level tasks. Some phonemic tasks may be completed by reference to alphabetic knowledge; it is relatively easy to decide that the word *cat* contains three phonemes if one knows that it is spelled with three letters. For rhyme-level tasks, however, it is less easy to use alphabetic knowledge in this way as it is necessary to know what it means for two words to rhyme.

In summary, evidence as to the nature of hyperlexic children's phonological-processing skills is equivocal. Despite their difficulties with language and communication, it is possible that their speech skills are relatively strong, and certainly, the fact that they have developed surprisingly good word recognition suggests their phonological skills should be adequate. Consistent with this, hyperlexic children perform well on tests such as phoneme deletion. In contrast, some hyperlexic children have greater difficulty with tests involving rhyme. Future experiments comparing the performance of hyperlexic children with appropriate controls while varying both the linguistic level (phoneme, syllable, rhyme) and the cognitive demands of phonological tasks (like those that have been conducted with normal children, e.g., Nation & Hulme, 1997) should elucidate the nature of hyperlexic children's phonological-processing skills. In the meantime, the observation that most hyperlexic children perform well on phoneme-level tasks suggests that their phonological skills are reasonably well developed and can support the learning of print–pronunciation associations.

Component Reading Skills in Hyperlexia

The nature of the underlying cognitive mechanisms that support hyperlexic children's precocious word recognition has typically been addressed with a standard dual-route framework. As described earlier, dual-route models propose that word recognition is accomplished by two main routes, a phonological route that uses grapheme–phoneme correspondence rules and a lexical route that activates word-specific representations. By comparing the reading of nonwords, regular, and exception words, investigators have examined whether hyperlexic reading is characterized by an overreliance on either the phonological or the lexical routines.

Nonword Reading Skills

A number of case reports and experimental studies provide evidence that hyperlexic children are able to read nonwords. Consequently, it has been argued that they have an adequate phonological reading route and that their reading precocity is not the result of an overlearned sight vocabulary (e.g., Aram et al., 1984; Glosser et al., 1996; Goldberg & Rothermel, 1984; Seymour & Evans, 1992; Siegel, 1984). Moreover, the nonword reading of a group of hyperlexic children studied by Snowling and Frith (1986) was equivalent to that of reading age (RA) control children, indicating that their decoding skills were perfectly in line with word-recognition level. Some hyperlexic children possess nonword decoding skills that are considerably in advance of their word-recognition skills (Aram, 1997). However, it is not clear whether all hyperlexic children show average or above average nonword reading skills. Aram and Healy (1988) described anecdotal evidence suggesting that some hyperlexics have great difficulty reading nonwords. Similarly, out of the 8 hyperlexic children tested by Goldberg and Rothermel (1984), 3 had difficulty reading even simple monosyllabic nonwords, as did DL, a hyperlexic teenager studied by Aaron, Frantz, and Manges (1990).

Although the weight of the evidence suggests that hyperlexic children read nonwords satisfactorily, a number of questions remain to be answered. In the normal population, children and adults find nonwords that are similar to familiar real words easier to read than nonwords that share fewer neighbors. For example, Treiman, Goswami, and Bruck (1990) reported nonwords sharing spelling-sound segments with many other words, that is, nonwords that have many neighbors (e.g., tain) were read with greater ease than nonwords sharing fewer real word neighbors (e.g., goach). Clearly, therefore, not all nonwords are equally easy to read, and it is possible that some of the variation between different studies describing hyperlexic children's nonword reading skills may be a consequence of different investigators using different types of stimuli. This possibility has not been investigated systematically, although Glosser et al. (1996) presented some relevant data from a retarded, hyperlexic 10-year-old boy. This child was as good as RA controls at reading nonwords that shared real-word analogies (e.g., nove), but he was much worse with items that were not word-like (e.g., fwov).

It is interesting to consider why non-word-like nonwords were more difficult for this hyperlexic child. Nonword reading requires a number of skills such as segmentation, verbal memory, and blending, and arguably, these component tasks may be much easier to perform when reading nonwords that are analogous to familiar real words. When an analogical strategy is not available, hyperlexic children may find nonword decoding more difficult than RA control children, especially when phonotactically illegal nonwords (i.e., items containing letter and sound sequences that do not occur in English) are used. Speculatively, it may be the case that hyperlexic children can decode words containing frequently occurring print–sound components but have greater difficulty reading nonwords containing less frequent units. This possibility needs to be investigated in a study comparing hyperlexic and RA control children's reading of nonwords that vary in neighborhood size and word likeness.

Regular and Exception Word Reading

An advantage for reading regular words over exception words has been taken as important evidence for the use of the phonological reading route. A small number of studies have investigated hyperlexic children's ability to read exception words. Welsh et al. (1987) and Aram et al. (1984) found that hyperlexic children made fewer errors reading regular words than exception words, though in each case, the children were able to read some exception words. Both studies concluded that hyperlexia is best characterized in terms of an intact phonological route accompanied by a partially impaired direct lexical route. A limitation of these studies was the lack of the control group of normal children reading at the same level. Indeed, Frith and Snowling (1983), who compared the performance of hyperlexic autistic children with younger RA control children, found the hyperlexic children's regularity effect comparable to that shown by the control group.

A different conclusion followed from a case study by Glosser et al. (1996), who compared a 10-year-old retarded hyperlexic child's reading with that of 10 control children reading at the same level on a standardized test of word recognition, the Wide Range Achievement Test (WRAT; Jastak & Wilkinson, 1984). Whereas the hyperlexic child read the same number of regular words correctly as the control children (84% and 85%, respectively), he read only 50% of the exception words compared with the control group's mean of 69%. Glosser et al. did not consider him to be impaired at reading exception words because his score fell less than two standard deviations below the control mean. This cutoff, however, seems overly rigorous. Given that the child was matched to the controls for overall WRAT reading level (and the WRAT contains both regular and irregular words), the finding that his exception word reading was more than one standard deviation below the controls is noteworthy. Although it is the case that he was able to read exception words, he appears to have been less accurate than RA control children.

In summary, previous investigations provide some evidence to suggest that hyperlexic children have relatively weak exception-word reading. However, it is inaccurate to categorize their reading as relying exclusively on the phonological reading route; this underscores the general observation that in development, sharp dissociations between word recognition for regular and exception words are rare (Snowling, Bryant, & Hulme, 1996).

Reading Comprehension

In contrast to a number of studies examining word recognition, surprisingly few have investigated hyperlexic children's reading comprehension systematically. One consistent finding to emerge is that hyperlexic children tend to perform equally poorly on tests of both reading comprehension and auditory comprehension (Goodman, 1972; Healy et al., 1982; Temple, 1990; Welsh et al., 1987). This suggests that their difficulties are not limited to written text processing but instead may be a reflection of underlying language-comprehension deficits. A related issue concerns whether or not comprehension is unexpectedly poor in this group. Welsh et al., using the RQ methodology described earlier, showed that for their hyperlexic sample, reading comprehension was not any lower than would be predicted from verbal mental age (MA). They argued that hyperlexic children do not have a specific reading comprehension deficit other than that accounted for by general language-comprehension difficulties, as measured by Verbal IQ.

Along with Welsh et al. (1987), a number of studies have found hyperlexic children to have impaired comprehension of single words (e.g., Aram et al., 1984; Goldberg & Rothermel, 1984; Healy et al., 1982) as measured by written word analogues of tests such as the Peabody Picture Vocabulary Test (PPVT) or the British Picture Vocabulary Scale (BPVS) (Seymour & Evans, 1992; Snowling & Frith, 1986). Weak local-level skills such as poor vocabulary are likely to compromise subsequent text-level comprehension. However, it is not the case that hyperlexic children show no comprehension of single words. Frith and Snowling (1983) used a Stroop paradigm in which children were shown a color name written in an ink of a conflicting color (e.g., the word red written in blue ink) and were asked to ignore the written word and name the color of the ink. Like normal controls, the hyperlexic children were slower to name the color when it conflicted with the written color name. This interference suggests that the hyperlexic children were able to access the meanings of written words.

Sentence-level reading comprehension was measured by Seymour and Evans (1992) using a sentence verification (true/false) task. The hyperlexic boy MP was completely unable to answer negative statements correctly (e.g., Cars are not animals; A dog can't run). These findings are consistent with Snowling and Frith's (1986) observation that hyperlexic children perform poorly on the Test for the Reception of Grammar (Bishop, 1983), regardless of whether it is presented in auditory or written word formats.

Snowling and Frith (1986) also reported a series of experiments designed to investigate sentence and text-level comprehension in hyperlexic children. They included an interesting comparison between high-verbal ability hyperlexic children (whose verbal mental age was consistent with reading age) and low-verbal ability hyperlexic children (verbal mental age less than reading age). To summarize their findings briefly, hyperlexic children were less good than RA control children at selecting appropriate words in a cloze task presented in the context of a story. For example, in the sentence "their *mother/friends/room* led them to the pond," both *mother* and *friends* were sentence-appropriate alternatives, but *mother* was also story-appropriate. Although the hyperlexic children could reject semantically anomalous distractors (e.g., *room*), they were unable to integrate information from the story as a whole to help them choose between story-appropriate and sentence-appropriate alternatives. Important, however, only the low-ability hyperlexic children were impaired on this task; the high-ability hyperlexic children performed equivalently to the RA controls.

Alert to the fact that the modified cloze procedure may have enhanced the hyperlexic children's comprehension, Snowling and Frith administered two further tests of reading comprehension. First, they asked the children to read a passage and simply cross out any anomalous words (e.g., The hedgehog could smell the scent of the *electric* flowers). In addition, they also asked a number of comprehension questions, some of which required verbatim memory for information presented in the story (e.g., For how long had the hedgehog been in her underground nest?), whereas others could also be answered by using general knowledge (e.g., What makes hedgehogs wake up from their winter sleep?). The low-ability hyperlexic children performed less-well than controls: They crossed out as many appropriate as inappropriate words and had great difficulty answering both factual and general knowledge questions. As before, however, the high-ability hyperlexic children performed as well as younger, RA control children.

There are many questions that remain to be answered concerning the comprehension skills of hyperlexic children. However, data from Snowling and Frith's study suggest that in addition to vocabulary and word-knowledge deficits, hyperlexic children are less able to engage in many of the processes necessary for successful comprehension: They process texts superficially, failing to process information at a story level, and they are less able to integrate new information with general knowledge when reading.

Early Onset and Unusual Preoccupation With Reading in Hyperlexia

A number of reports describe how many hyperlexic children begin to read in the preschool years, without any special tuition or obvious instruction (Aram et al., 1984; Goldberg & Rothermel, 1984; Healy et al., 1982; Mehegan & Dreifuss, 1972). Indeed, some experimental studies have only included hyperlexic children with a preschool history of hyperlexia (Aram et al., 1984). However, as Aram and Healy (1988) pointed out, it is not clear whether early onset ought to be a defining feature of hyperlexia, and if so, how early it must emerge for it to be clinically significant. Additionally, as hyperlexia is often associated with serious developmental delay, early onset relative to other cognitive abilities may be more relevant than early onset relative to chronological age. To address this issue satisfactorily, it would be necessary to compare hyperlexic children with a history of early onset with similarly hyperlexic children for whom there is no evidence of early onset. Only if substantial quantitative or qualitative differences emerged between the two groups would it be necessary to include early onset as a defining feature of hyperlexia.

Another feature of hyperlexia that has been reported frequently is the strong preoccupation with reading that many hyperlexic children show. Healy et al. (1982) described how reading replaced other play activities in their sample of hyperlexic children. Potentially, the fact that hyperlexic children are compulsively preoccupied with print (and therefore receive extensive reading experience and practice) may provide important information concerning how and why they develop surprisingly advanced word-recognition skills.

Although this issue has not been systematically examined in hyperlexia, it has been investigated in individuals possessing a variety of other savant characteristics. The term *savant* is used to describe mentally retarded individuals who show exceptional ability in an isolated area, such as the ability to calculate calendrical information, or exceptional musical, artistic, or memory skills (for review, see O'Connor & Hermelin, 1988). As in hyperlexia, savant skills emerge at an early age without any specific instruction. In addition to exceptional rote-memory skills (Nettlebeck & Young, 1996; O'Connor & Hermelin, 1989), savant abilities are associated with considerable practice and preoccupation with their skill over a long period of time. This is consistent with the anecdotal reports describing the early onset and extensive preoccupation with reading seen in hyperlexia.

One possibility is that such traits are more a symptom of autism rather than the savant skill itself. However, as is discussed in greater detail later, many hyperlexic children are also autistic or show autistic-like behaviors. Important, although most individuals showing savant characteristics are autistic, O'Connor and Hermelin (1991) found that both autistic and nonautistic savants showed more preoccupations, compulsive behaviors, and repetitive inclinations than controls. Thus, obsessional preoccupation with one activity is related to savant abilities, regardless of whether or not an individual is autistic.

Turning to the question of whether excessive preoccupation with reading is fundamental to hyperlexia, O'Connor and Hermelin (1991) concluded that the tendency for savants to be preoccupied with one area, and to practice their skill excessively, is a necessary but not sufficient condition for the manifestation of high-level performance in people with limited cognitive ability. On this view, excessive practice in combination with particular patterns of other strengths and weaknesses are the essential ingredients needed for a hyperlexic reading system to develop.

What this combination of strengths and weaknesses may be is discussed later; at the present time, however, two conclusions may be drawn. First, the obsessional attitude to reading shown by hyperlexic children may be a vital component of the disorder; if children are to develop hyperlexia, they need to devote considerable attention and practice to the task of reading. Second, the frequent observation that many children are autistic or show autism-spectrum characteristics is consistent with the fact that ritualistic and obsessional behavior is a key feature of autism.

Relationship of Hyperlexia With Other Developmental Disorders

Autism

Throughout the extant literature, there is a well-recognized relationship between autism and hyperlexia. Relative strengths in rote memory and perceptual processing along with relative weaknesses in associative skills and reasoning are strikingly similar to the cognitive profile often seen in autistic individuals (e.g., Happé, 1994). As described above, the obsessional preoccupations that accompany hyperlexic reading are consistent with the pattern of obsessional, repetitive, and ritualistic behavior associated with autism (Tirosh & Canby, 1993). Moreover, the speech, language, and communication difficulties inherent in hyperlexia are the same as those typically reported in autism-spectrum disorders (Aram & Healy, 1988). Important, however, hyperlexia is not specific to autism; hyperlexic individuals with autism show qualitatively the same patterns of reading behavior as hyperlexic individuals without autism (Snowling & Frith, 1986).

Nonetheless, as hyperlexia is so often a feature of autism, it is useful to consider its cognitive characteristics and how these may predispose individuals to exceptional reading ability. One view of the cognitive deficit underlying autism is that autistic children are poor at understanding and attributing mental states (e.g., Baron-Cohen, Leslie, & Frith, 1985). Despite considerable evidence for this theory, it does not explain a number of features that may well be relevant to the development of hyperlexia, namely, circumscribed interests and savant skills. To provide an explanation for such symptoms, Frith and colleagues (1989; Frith & Happé, 1994) have suggested that an additional cognitive characteristic of autistic individuals is weak central coherence. Central coherence refers to the strong tendency to interpret information with reference to context. Thus, autistic individuals who show weak central coherence are poor at using context to construct meaning; rather than analyzing information globally (attending to the whole), they analyze in a piecemeal fashion (attending to the parts). It can be seen that in relation to reading, this cognitive style may lead to a preoccupation with decoding rather than with global meaning. Consistent with this theory, autistic individuals perform relatively well on Block Design and Embedded Figures as they are able to resist the overall pattern and concentrate instead on the constituent parts (Shah & Frith, 1993). Likewise, autistic individuals are no better at recalling lists of related words than lists containing unrelated items (Tager-Flusberg, 1991). Thus, they are unable to use meaning to integrate information to aid recall.

Happé (1997) investigated the possible relationship between central coherence and reading skills in autistic children using a task in which participants read aloud words presented in context. She compared their pronunciation of homographs (words that are spelled the same but have different pronunciations depending on meaning) in sentences such as "There was a big *tear* in her dress" and "The girls were climbing over the hedge. Mary's dress remained spotless but in Lucy's dress there was a big *tear*." In the first sentence, the homograph appears before any disambiguating context, whereas in the second, the homograph occurs later in the sentence when the appropriate meaning (and therefore pronunciation) should be available. Autistic children (even those who performed well on theory of mind tasks) made very little use of preceding sentence context, and they were far less likely to produce context-appropriate pronunciations. Happé (1997) interpreted this as a difficulty in extracting meaning from context because of weak central coherence.

The sentences used by Happé (1997) were originally used by Snowling and Frith (1986). The results of their study are important as they show that nonautistic hyperlexic children performed in the same way as the autistic children. Thus, one possibility is that weak central coherence is a cognitive style common to individuals who show hyperlexia. According to this view, inflexible reading and lack of sensitivity to contextual constraints may be just one facet of an underlying deficit in central coherence. This evidence suggests that hyperlexic children may have the kind of cognitive style characterized by weak central coherence. Relating back to the task analysis of reading described earlier, although weak central coherence would assist children with learning print-pronunciation associations (attending to the parts), it would seriously hinder their ability to perform many of the constructive and integrative processes needed for successful comprehension.

In summary, hyperlexic children—even those hyperlexic children who are not autistic—show characteristics typically associated with autism. Weak central coherence coupled with an obsessive preoccupation with reading provide an ideal learning environment for the development of excellent decoding skills for those children who have reasonable phonological, orthographic, and associative memory skills. Thus, these two characteristics may be potentially important in explaining hyperlexic children's reading behavior.

Dyslexia

The classic discrepancy approach to defining dyslexia views dyslexia as specific reading difficulties in children with otherwise normal cognitive abilities, and currently accepted definitions of dyslexia (e.g., Orton Dyslexia Society, 1994; Shaywitz, 1996) stress inferior single-word recognition. In contrast, hyperlexia refers to superior word recognition in children who typically have some degree of cognitive disorder. From this description (see also Aaron, 1989; Aaron et al., 1990; Seymour & Evans, 1992), dyslexia and hyperlexia appear to be two clearly separate and opposing disorders. Indeed, Snowling (1987) commented on the opposing strengths and weaknesses of dyslexic and hyperlexic children when reading stories taken from the Neale Analysis of Reading Ability (Neale, 1989). Whereas the dyslexic children were slow, hesitant, and error prone, the hyperlexic children read fluently and accurately. Yet, despite their difficulties with the mechanics of reading, the dyslexic children exhibited far superior comprehension than the hyperlexic children.

However, a number of studies have made the claim that hyperlexia is a variant of developmental dyslexia, and in at least one classification system, hyperlexia is defined as a subtype of dyslexia (Child Neurology Society, 1981). de Hirsch (1971) made a strong claim that hyperlexic children should in fact be considered as dyslexic. Her rationale for this was that the exceptional word recognition seen in hyperlexia is in no sense real reading, that is, reading with satisfactory comprehension. Instead, she viewed hyperlexic reading as mere word calling and concluded that this was a symptom of reading disability.

Evidence from family studies has also been used to support the notion that dyslexia and hyperlexia are related conditions. Healy and Aram (1986) found that 11 out of the 12 hyperlexic children they studied had a family history of language, reading, or spelling problems or a learning difficulty of some description, particularly through the paternal line. On the basis of these essentially descriptive data, however, it is premature to make any conclusions concerning the potential overlap between the two disorders. It is worth noting that Healy and Aram's study was based entirely on family questionnaire and interview data from a very small population sample; no objective tests of reading, language, or IQ were administered to family members. Thus, it is not clear whether any of the family members were truly dyslexic or whether their literacy difficulties were a consequence of more general cognitive/language impairments or even an over-reporting of symptoms in clinical families.

Although it is true to say that hyperlexia is a reading disorder (at least a disorder in terms of reading comprehension), it also seems very clear that hyperlexia is not equivalent to dyslexia. Although many dyslexic children do have subtle language problems (for a recent review, see Snowling & Stackhouse, 1996), it is well established that primary impairments in phonological processing underpin their difficulties with word recognition (Brady & Shankweiler, 1991; Hulme & Snowling, 1992; Snowling, 1991; Stanovich & Siegel, 1994). In contrast, evidence reviewed earlier suggests that hyperlexic children have normal phoneme-level phonological-processing skills despite pervasive cognitive and language impairments.

In summary, evidence that hyperlexia is a variant of dyslexia is clearly lacking, and instead, the available evidence is more consistent with the view that dyslexia is an opposing disorder to hyperlexia. As suggested recently by Cohen et al. (1997), hyperlexia should be viewed as a variant of a more general language impairment rather than a variant of developmental dyslexia. Although the observation that dyslexia and hyperlexia co-occur in families is interesting and certainly worthy of further investigation, suffice it to say that the cognitive and behavioral manifestations of the two disorders are very different, and as such, it is appropriate to consider hyperlexia and dyslexia as two distinct developmental disorders.

Children With Chromosomal Abnormalities

Turner's Syndrome (TS) is a disorder caused by an abnormality on the short arm of the second X-chromosome. TS individuals typically have normal verbal skills relative to impaired nonverbal and spatial skills (Shaffer, 1962). Temple and Carney (1996) examined the reading skills of 29 girls with TS. They found that their word recognition and reading comprehension were significantly higher than CA controls, and they were better at reading both irregular words and nonwords. These findings led Temple and Carney to conclude that "Hyperlexia is not therefore a necessary result of a disordered language system but may emerge in association with strong language skills" (Temple & Carney, 1996, p. 343). However, to label these children as hyperlexic is misleading: As the TS children showed superior reading accuracy and comprehension abilities, they are in essence generally good readers rather than hyperlexic. Thus, it is more appropriate to conclude that the TS girls studied by Temple and Carney are not hyperlexic but instead have generally good reading skills.

Rovet et al. (1995) reported the case of a 4-year-old boy with a chromosome abnormality involving deletion of the short-arm of chromosome 20. Despite delayed speech, language, and motor development, decoding and comprehension skills were extremely advanced. Rovet et al. described this child as having savant characteristics. Given that his reading-comprehension skills were also in advance of the levels predicted by his IQ, this term, rather than hyperlexia, is a more accurate description of his reading behavior.

Finally, it is worth noting that children with Down syndrome often show word-recognition skills that are in advance of mental age, and therefore, they too might be considered to be hyperlexic. A group of 30 children with Down syndrome studied by Mercer (1997) achieved a mean reading age of 6 years 3 months, whereas their mean verbal MA was only 4 years 3 months. Similarly, Cossu and Marshall (1990) described an 8-year-old boy with Down syndrome who was reportedly the best reader in his class, despite a verbal MA of only 4 years 6 months. As is the case in hyperlexia, word recognition often exceeds language comprehension, although reading-comprehension skills are in line with language-comprehension skills (Fowler, 1995). Interestingly, however, Down syndrome children are seldom referred to as hyperlexic. This may be a consequence of the fact that although their word recognition is well in advance of MA, it tends to be far below age-expected levels. In Mercer's study, for example, the Down syndrome children had a mean chronological age of 13 years 2 months. In contrast, in Welsh et al.'s (1987) study, 4 out of 5 of the hyperlexic children demonstrated word-recognition skills far in advance of the levels expected for their chronological age. It can be inferred that a large deficit in reading age relative to chronological age precludes use of the term hyperlexia, even in cases where a child's word recognition is considerably in advance of language-comprehension skills.

Precocious Readers

Pennington et al. (1987) described the very precocious reading skills of a preschool child whose development was normal in all other respects (no signs of autistic-like behavior, language, communication, or motor difficulties). When tested at ages 2 years 11 months and 4 years 2 months, he showed word-recognition skills some 6 to 7 years above his age level. As with hyperlexic children, his reading skills developed spontaneously without special instruction. Using an RQ index, Pennington et al. showed that this level of word recognition was well in advance of the levels predicted by his (very high) IQ. He was able to read nonwords, even those with complex letter–sound correspondences, and he was good at exception word reading. In sum, his exceptional word recognition represented "a normal balance between rule-like and word-specific strategies" (Pennington et al., 1987, p. 172). This child also had good reading comprehension, consistent with the level predicted by his superior VIQ and, arguably, limited only by his oral receptive language.

Thus, extreme and unexpected word-recognition precocity can occur in normally developing children who have no language or social impairments. It follows that the extraordinary word-recognition skills seen in hyperlexia may not necessarily be a consequence of language disorder. Importantly, however, normal precocious readers differ from hyperlexic children in that their comprehension is also in advance of age-expected levels; despite showing advanced word-recognition skills, normal precocious readers are not hyperlexic as they are able to understand much of what they read.

Children With Comprehension Difficulties

Approximately 10–15% of 7–9-year-old children have been reported to have reading-comprehension difficulties in the face of normal word recognition (Nation & Snowling, 1997; Stothard & Hulme, 1992; Yuill & Oakhill, 1991). These children show normal reading accuracy but have difficulty understanding what they read. Although they show normal range phonological-processing skills, they have impaired listening comprehension and below-average vocabulary and grammatical understanding (Nation & Snowling, 1998a; Stothard & Hulme, 1992, 1995). They are also poor at making inferences, and they have a variety of metacognitive weaknesses such as a tendency to fail to monitor comprehension closely (Yuill & Oakhill, 1991). Given these findings, it is appropriate to view poor comprehenders' reading comprehension difficulties within the context of their more general spoken language impairments, rather than as specific difficulties with reading comprehension. Thus, in many senses, poor comprehenders show a similar style of reading to hyperlexic children (good word recognition but poor comprehension) despite forming part of the normal, nonclinical population.

Investigations of poor comprehenders' reading skills also support the view that they share a similar profile to that typically seen in hyperlexia, albeit less severe. Despite having good nonword reading and age-appropriate reading accuracy, Nation and Snowling (1998b) found that when faced with a word they were unable to decode, poor comprehenders were less adept at using contextual information to facilitate word recognition. Moreover, although their speed and accuracy of reading high frequency and regular words was equivalent to control children, they were slower and more error prone at reading words that could not be read so easily using a decoding strategy, namely, low frequency and exception words (Nation & Snowling, 1998a). These results are similar to those showing reduced context effects (Snowling & Frith, 1986) and relatively weaker exception word reading (Aram et al., 1984; Welsh et al., 1987) in hyperlexic children.

To date there is no evidence to suggest that poor comprehenders share the same compulsive preoccupation with reading as hyperlexic children, or that they begin to read at a very young age. Yet, although poor comprehenders show less severe deficits, they are similar to hyperlexic children in terms of the pattern of strengths and weaknesses of their reading skills.

Toward a Definition of Hyperlexia

Diagnosing Hyperlexia

Clearly, hyperlexic children have exceptional word recognition; this is well illustrated when an RQ index is used to predict word recognition from IQ (Welsh et al., 1987). However, a number of studies have shown that their reading-comprehension skills are entirely consistent with verbal mental age: When an RQ index is computed predicting reading comprehension from verbal ability, hyperlexic children do not show a comprehension deficit. Thus, although their word recognition is surprisingly advanced, their comprehension is not surprisingly poor. This observation raises the issue of whether or not such children are truly hyperlexic. Snowling and Frith (1986) (see also Rispens & Van Berckelaer, 1989) recommended that the label hyperlexia should be reserved for those individuals whose word-recognition skills are surprisingly advanced and whose comprehension is surprisingly poor, that is, worse than predicted by verbal mental age.

If Snowling and Frith's strict definition of hyperlexia were to be adhered to, most of the children described in the literature would no longer meet the criteria to be classified as hyperlexic as their reading comprehension is not impaired relative to mental age (Healy et al., 1982; McClure & Hynd, 1983; Welsh et al., 1987). This is not surprising given the fact that comprehension skills are inherently related to general verbal ability (Sternberg, 1987). According to this view, children with good verbal skills learn the meanings of words more quickly, and they are able to access word meanings more rapidly. Weak verbal skills hinder the acquisition of word meanings, and consequently, this impoverished word knowledge will impede comprehension. Given this intimate relationship between comprehension and verbal ability, it is not surprising that few hyperlexic children show comprehension skills substantially below what would be predicted from MA. Consequently, Snowling and Frith's suggestion that only those children who have lower than expected comprehension be classed as hyperlexic is over strict and untenable; adopting such stringent diagnostic criteria risks failing to identify those children who have exceptional word recognition but do not show striking comprehension deficits relative to MA.

At the very least, however, a diagnosis of hyperlexia should only be applied if a child's comprehension skills fall well below the level predicted from both their word-recognition level and chronological age. Without this limitation, there is a danger that children who are essentially normal readers will be described as hyperlexic. This danger is well illustrated by Temple's (1990) case study of MS, a 10-year-old boy diagnosed as hyperlexic on the basis of his performance on the Neale Analysis of Reading Ability (Neale, 1989). In this test, children read aloud short passages of text and then are asked a set of comprehension questions. Although he achieved a reading accuracy age of 12 years 4 months, his reading comprehension age was only 9 years 8 months. Using a regression technique provided by Yule, Lansdown, and Urbanowicz (1982), Temple demonstrated that his reading accuracy was 2 years 4 months higher than would be predicted from chronological age and IQ. In comparison, his reading comprehension was only 3 months higher than predicted from chronological age and IQ.

Findings from a large-scale study of 355 8–10-year-old children (Nation & Snowling, unpublished data) suggest that Temple's description of this child as hyperlexic is inappropriate. Of those children with average (for age) reading comprehension as measured by the Neale Analysis of Reading Ability (children like MS), some 32% had a reading accuracy age at least 6 months higher than their comprehension, and 12% had an accuracy age over 2 years in advance of their comprehension skills. Thus, the profile of the child studied by Temple is not particularly unusual or atypical. Moreover, it is inappropriate to classify a child such as MS as hyperlexic, or even as a poor comprehender, as his comprehension was perfectly average for his chronological age.

In summary, precise definitions of hyperlexia are lacking. It is clear that to be described as hyperlexic, a child should show significantly advanced word recognition (e.g., as measured by an RQ index). In addition, hyperlexic children should have reading comprehension that is significantly impaired with respect to both word-recognition level and chronological age. Those children who also have impaired reading comprehension relative to verbal mental age appear to be very rare; this is not surprising given the intimate relationship between comprehension and verbal ability (e.g., Sternberg, 1987).

Syndrome or Symptom?

Part of the confusion surrounding what is and what is not hyperlexia arises from the question of whether hyperlexia constitutes a syndrome. A syndrome may be defined as a cluster of symptoms that occur together and that, taken together, are indicative of a particular disease or abnormality (e.g., Reber, 1985). Advocates of the syndrome view (e.g., Aaron, 1989; Healy, 1982) view hyperlexia as a primary disorder consisting of three symptoms, namely, impaired comprehension, exceptional word recognition, and the spontaneous development of reading before 5 years of age. On this view, hyperlexia is a clearly defined syndrome, separate and distinct from other disorders.

However, it remains equivocal whether or not hyperlexia constitutes a syndrome. One limitation of studies that offer support for the syndrome view is that the selection criteria used would have precluded children who did not show the defining features of hyperlexia from taking part. For example, criteria for inclusion in Healy et al.'s (1982) study included intense interest in graphic symbols, onset of reading prior to 5 years of age, disordered language and cognitive functioning, and word recognition greatly in advance of other linguistic and cognitive abilities. As all 12 of the hyperlexic children studied showed these symptoms, it was concluded that these features are key features of a hyperlexic syndrome. Clearly, however, this is circular and difficult to interpret; although impaired comprehension and advanced word recognition are defining features of the disorder, it is questionable whether the very early onset of reading development needs to be present for a child to be described as hyperlexic.

Perhaps more important, the term *syndrome* implies that the co-occurring symptoms stem from the same underlying cause and that this cause is necessary and sufficient to explain the presence of the disorder. In cases of (appropriately defined) hyperlexia reported in the literature, hyperlexic reading is comorbid with other disorders, and many of the behaviors shown by hyperlexic children are also defining characteristics of language impairment, mental retardation, or autism. On this view, hyperlexic reading behavior is a symptom of an underlying disorder, rather than being the primary disorder itself. The one characteristic that does differentiate hyperlexic children is their superior word recognition, and this characteristic also can be found in otherwise normal children (Pennington et al., 1987). McClure and Hynd (1983) concluded that hyperlexia should be considered a "secondary symptom" that sometimes occurs in children who are otherwise cognitively, linguistically, or socially impaired. The weight of evidence is consistent with this view, namely, that hyperlexia is a behavioral characteristic rather than a diagnostic category, and as such, it is appropriate that it does not appear as a diagnostic entry in classifications such as the fourth edition of the *Diagnostic and Statisti-*

cal Manual of Mental Disorders (American Psychiatric Association, 1994).

Regardless of whether or not hyperlexia is considered a syndrome, what remains to be explained is how such exceptional reading skills develop. In the remainder of this review, we argue that hyperlexia is best understood by considering individual differences in the interplay between children's strengths and weaknesses across a variety of reading-related areas. Before discussing hyperlexic word recognition within such a theoretical framework, it is instructive to summarize the key features and characteristics of the disorder:

1. Objective measures such as RQ indexes show that for most hyperlexic individuals, word recognition is exceptionally advanced with respect to MA and often with respect to CA.

2. In most cases, reading comprehension is commensurate with the levels expected for MA, although there may be a minority of children for whom reading comprehension is also impaired relative to MA.

3. Most hyperlexic individuals show (sometimes profound) cognitive, linguistic, and social impairments. Verbal skills are usually impaired, and there is a tendency for poor performance on tasks requiring problem solving, decision making, or judgment. In contrast, rote memory and perceptual abilities are relative strengths.

4. Descriptions of hyperlexic children's word recognition reported in the literature point to considerable variation and heterogeneity. It is not the case that word recognition in hyperlexia is mediated by over-reliance on visual or phonological strategies, although evidence suggests that hyperlexic children are relatively weak at reading exception words. Generally, hyperlexic children have been found to have good nonword reading, although a small number of case studies describe hyperlexic children with very poor nonword reading abilities.

5. Hyperlexia is not a variant of developmental dyslexia; in dyslexia, word recognition is surprisingly poor, whereas in hyperlexia, word recognition is surprisingly good.

6. Many hyperlexic children are autistic or show behavior that is described as autistic-like. Two features of autism, preoccupation with circumscribed interests and weak central coherence, may be particularly relevant for understanding the pattern of reading behavior associated with hyperlexia.

7. Only children who have exceptional word-recognition skills and impaired comprehension skills (relative to chronological age) should be described as hyperlexic. The available evidence suggests that hyperlexia is not a discrete disorder; rather, it is a pattern of reading behavior that is sometimes observed in children with cognitive, linguistic, and social handicaps.

Toward a Theoretical Framework

As is clear from the literature review, reading comprehension is generally poor in hyperlexia. Although much work needs to be done to fully understand the nature of hyperlexic children's reading comprehension, their deficit in comprehension is usually commensurate with difficulties in comprehending spoken language. What seems paradoxical in hyperlexia is the fact that cognitively and linguistically impaired children are able to develop unexpectedly strong word-recognition skills. This final section addresses this issue by considering hyperlexic children's word recognition within a framework of normal reading development.

The Importance of Individual Differences

If hyperlexia is to inform our understanding of the nature of normal reading development, it is important to be able to explain hyperlexic reading within a theoretical model of normal reading development. By recognizing that it is a pattern of reading behavior, hyperlexia can be thought of as part of the normal variation in reading skills. This is illustrated in Figure 1. The *x*-axis represents variation in word-recognition skills moving left to right from poor to exceptional. Similarly, the *y*-axis represents individual differences in language-comprehension skills moving from top (exceptional) to bottom (poor). Whereas dyslexic children have adequate language comprehension but poor word recognition (Snowling, 1987), hyperlexic children show acceptable word recognition but impaired comprehension. Also shown in Figure 1 are children with specific comprehension difficulties; as discussed above, these children present a similar profile to that seen in hyperlexia but to a less severe degree. Normal precocious readers who have exceptional word recognition and comprehension fall within the upper right quadrant of Figure 1. Such children may be compared with children in the lower left quadrant who have generally poor reading skills (i.e., impaired comprehension and impaired word recognition).

Very clearly, word recognition and comprehension are complex sets of skills that depend on many different processes. Given the strong relationship between phonological-processing skills and word recognition, it is tempting to suggest that individual differences in word recognition are largely accounted for by differences in phonological skills, whereas individual differences in comprehension may be more a consequence of differences in semantic and syntactic processing skills. Ultimately, it is important to remember that the two sets of skills are not fully independent as Figure 1 would suggest; word recognition plays a role in reading comprehension, and language-comprehension skills also influence the development of skilled word recognition (Nation & Snowling, 1997). For present purposes, however, Figure 1 serves as a useful

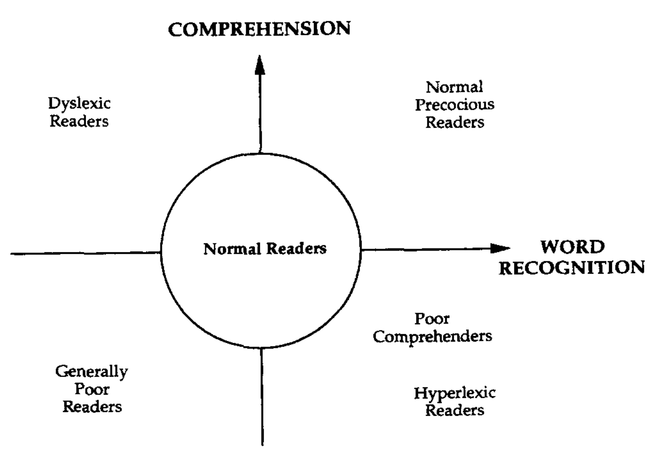

Figure 1. Schematic framework showing hyperlexia as a part of the normal variation in reading skills.

schematic framework for considering individual differences in reading development.

An Alternative Theoretical Framework

One way to think about how these individual differences impact reading development is from the perspective offered by connectionist modelling (see Plunkett, Karmiloff-Smith, Bates, Elman, & Johnson, 1997, for a detailed review of the importance of connectionism to developmental psychology). As described earlier, connectionist models consist of many simple processing input and output units that are linked together via weighted connections. With experience, such models learn the statistical relationship between input and output patterns that exist in the training environment.

Plaut et al. (1996) described a connectionist model of normal reading development. They proposed that reading development may be best characterized as a division of labor between a phonological pathway and a semantic pathway (although two processes or pathways are implemented, this model is not akin to dual-route models as it does not use rules or have stored lexical representations; see Plaut et al. for discussion). The phonological pathway consists of connections between distributed orthographic and phonological representations. The semantic pathway deals with mappings between semantic, phonological, and orthographic representations. According to Plaut et al., in the early stages of learning to read, resources are devoted to establishing connections between orthography and phonology. As learning proceeds, semantic factors become increasingly important, particularly for the reading of exception words as these words are not handled well by the phonological pathway.

Word Recognition in Hyperlexia: A Connectionist Perspective

Earlier, evidence concerning hyperlexic children's reading of nonwords, regular words, and exception words was reviewed. It was argued that hyperlexic reading is not characterized by exclusive use of either the lexical or phonological route; it is not the case that hyperlexics only read via a phonological route as they are able to read some exception words, and the fact that many hyperlexics can read nonwords is support for the view that they do not rely exclusively on a lexical route. Within an interactive framework, however, the notion of distinct and separate routines is eschewed. In a connectionist model such as Plaut et al.'s (1996), reading behavior is simply a product of the architectural and computational properties of the network and the nature of the learning environment that the network has been exposed to. By analogy, an individual child's reading behavior emerges as a product of individual differences in the skills that they bring to the task of learning to read (e.g., quality of phonological or semantic representations) and their experience and familiarity with written words.

Considering hyperlexia within Plaut et al.'s (1996) framework, evidence suggests that hyperlexic children should be able to set up adequate connections between orthography and phonology. As hyperlexic children have well-documented difficulties with language and comprehension, however, they are more likely to have difficulties developing adequate connections between semantic, phonological, and orthographic representations. This will have few serious consequences for the reading of regular words or simple nonwords. In contrast, hyperlexic children may have greater-than-average difficulty reading exception words as these words are, according to Plaut et al., typically read with support from semantics. Consider the difficulty facing children learning to read words such as *chaos, enough,* or *unique,* if they have little understanding of the words' meanings.

A second consequence of a weak semantic pathway is that contextual cues will be less likely to facilitate word recognition compared with individuals whose reading systems are underpinned by adequate semantic skills. Nation and Snowling (1998b) have argued that sentence context may boost semantic input to the phonological units. When faced with an unfamiliar word, the combination of information from a partial decoding attempt with that gleaned from higher-level syntactic and semantic cues (in short, interactions between the phonological and semantic pathway) may allow the normal reader to read the word correctly. However, as a consequence of poor semantic skills, the word recognition of poor comprehenders (Nation & Snowling, 1998a) and hyperlexic children (Happé, 1997; Snowling & Frith, 1986) benefits very little from supportive context.

Another skill that may be influenced by a weak semantic pathway is verbal short-term memory. Patterson and Hodges (1994) studied 3 patients with progressive semantic dementia. These patients have general semantic deficits resulting in profound comprehension impairments. Interestingly, their recall for words they no longer comprehended was considerably worse than memory for lists of known words. Patterson and Hodges argued that semantic knowledge contributes to normal phonological processing (as indexed by a verbal short-term memory task) and that when this semantic support is not available, performance on a task such as verbal recall declines.

Although hyperlexic children generally have good memory for unrelated lists of words or digits (for review, see Aram & Healy, 1988), it may be that they are less able to use structure or meaning to bolster memory. Certainly, evidence suggests that this is the case in autism; for example, Tager-Flusberg (1991) compared autistic and MA control children's memory for semantically related and unrelated lists of words. Whereas the control children were better at remembering the related lists, the autistic children showed equivalent memory for both lists of words (see also Hermelin & O'Connor, 1967). As with their failure to use contextual cues to support word recognition, their poor semantic skills may result in their failing to use meaning-based structure to supplement verbal short-term memory.

An important characteristic of connectionist models is that learning is frequency sensitive. With enough experience, Plaut et al.'s (1996) implemented model was able to learn the correct pronunciation of most exception words, even without semantic support. The implication of this for hyperlexia is that children will be able to read some exceptions if they have been exposed to them a number of times. Given the well-documented preoccupation that hyperlexic children have with reading described earlier, it can be assumed that their exposure to words will be high, certainly higher than that of nonhyperlexic children of a similar mental age.

Glosser, Grugan, and Friedman (1997) provide some data that serves as preliminary support for this view. They were interested in a hyperlexic child's spelling of exception words and found that he was able to spell 67% of exception words correctly if he also knew the words' meanings, whereas for unknown words, he was able to spell only 14% correctly. After a period of intense training in which he was taught how to spell the same sets of words, he was able to spell 88% of known words and 71% of unknown words correctly. This indicates very clearly that it is possible to learn the spelling (and presumably also the pronunciation) of exception words without any knowledge of word meaning, provided that they have been encountered enough times.

Why Are Some Children Hyperlexic?

A theory of normal reading development in which both semantic and phonological processes influence word-recognition development seems an appropriate framework in which to consider word-recognition skills in hyperlexia. If it is accepted that individual differences in the balance of strengths and weaknesses govern the way in which the reading system develops, it is possible to make predictions concerning those characteristics a child would need to develop a hyperlexic reading profile.

Orthography and phonology. If children are to set up an adequate phonological pathway, they need to have sufficiently well-specified orthographic and phonological representations. The available evidence suggests that hyperlexic children have normal orthographic representations (e.g., they can still read words despite alterations in case or orientation; Goldberg & Rothermel, 1984). The evidence concerning their phonological-processing skills is less clear. Arguably, however, a reasonable prediction would be that although hyperlexic children may have difficulty with some phonological-awareness tasks (e.g., rhyme awareness), their underlying phonological representations are adequate. This prediction is consistent with the suggestion that hyperlexic children's speech skills are relatively strong. Indeed, this may set hyperlexic children aside from other language-impaired nonhyperlexic children who generally perform poorly on tests such as nonword repetition and phoneme deletion, even when their obvious language impairments have largely been resolved (Stothard, Snowling, Bishop, Chipchase, & Kaplan, 1998).

Semantics. Failure to develop an adequate semantic pathway may be a consequence of weak semantic skills. All reports of hyperlexia describe at least some degree of semantic impairment (e.g., poor vocabulary), and even for those hyperlexic children with normal range IQ (Richman & Kitchell, 1981), relatively weak verbal skills were noted. A likely consequence of this result is that their reading is unlikely to be underpinned by an adequate semantic pathway. In this respect, hyperlexic children differ from normal precocious readers whose word recognition is well supported by competent phonological and semantic pathways (Pennington et al., 1987).

Print exposure. Plaut et al.'s (1996) model learned the relationships between print and sound by extracting statistical regularities that occurred in the vocabulary to which the network had been exposed. Thus, another crucial variable is the nature of the written vocabulary that hyperlexic children have experienced. Earlier, it was suggested that children's preoccupations with reading may be a vital element of hyperlexia. Although no study has provided detailed information or presented objective estimates of print exposure (i.e., a child's familiarity and experience of written words), it is reasonable to predict that print exposure will be an important determiner of hyperlexics' word recognition (particularly of exception words), as it is in the normal population (Cunningham & Stanovich, 1991). Arguably, if a child with semantic and comprehension deficits is to develop advanced word recognition, they need to develop a strong phonological pathway. Without the privilege of support from semantics, this development requires considerable print experience. Those children who have autistic or autistic-like compulsions with reading will be more likely to become hyperlexic than other language-impaired children simply because they read more, and as a consequence, they are more sensitive to the relationship between print and pronunciation.

Memory. Verbal short-term memory skills are important for the development of word recognition, not least to enable children to blend and segment words (Wagner & Torgeson, 1987). Therefore, it is reasonable to predict that if a child is to develop a hyperlexic reading style, he or she would need to have good verbal short-term memory. The observation that hyperlexic children have relative strengths in tasks such as Digit Span (e.g., Aram et al., 1984; Cobrinik, 1974; Goodman, 1972; Richman & Kitchell, 1981) provides considerable support for this proposal. This observation is in stark contrast to the majority of specifically language-impaired children who show striking short-term memory deficits (e.g., Kamhi, Catts, Mauer, Apel, & Gentry, 1988). Thus, good short-term memory may differentiate language-impaired hyperlexic children from other language-impaired children who are not hyperlexic.

In addition to short-term memory, automatic word recognition requires print–sound associations to be rapidly retrieved from long-term memory. There is evidence to suggest that hyperlexic children possess strong long-term memory skills; indeed, a number of individuals with hyperlexia have also been described as savant mnemonists (e.g., Patti & Lupinetti, 1993). Important, however, their strengths are in rote memory. Although this will assist the learning and recall of print–sound associations, it will offer little support to the integrative and constructive processes required for text comprehension.

In conclusion, considering hyperlexia within this theoretical framework, the levels of word recognition observed in hyperlexia are the result of relative strengths (in phonological and orthographic skills, rote memory), relative weaknesses (in semantic skills), and exceptional levels of practice and exposure to print. On this view, children who develop a hyperlexic reading style have a particular profile of cognitive and linguistic abilities, coupled with considerable reading experience, plausibly resulting from narrow-focused preoccupations with reading.

Hyperlexia and Development

As with other disorders of development (specific language impairment, see Bishop, 1998; dyslexia, see Snowling et al., 1996), it is important that hyperlexia be studied within a developmental framework. Unlike studies of adult-acquired disorders, clinical studies of children are complicated by developmental factors. Claims such as those made by Glosser et al. (1997, p. 236) that a 10-year-old hyperlexic child has semantic deficits that are "stable and relatively circumscribed" need to be interpreted cautiously. Moreover, it is not just developmental change in an isolated clinical case that is relevant but also change and variation within the normal population. Hence, it is important that future studies compare hyperlexic children with appropriate control groups.

Depending on the particular hypothesis to be tested, researchers may wish to use control groups matched for mental age, reading age, language skills, or chronological age. To take the nature of phonological-processing skills in hyperlexia as an example, a comparison between SLI children with and without hyperlexia may provide evidence as to whether hyperlexic children have phonological skills that are greater than other children whose language comprehension and expression is similarly impaired. In addition, a comparison with mental age controls is also important as quite possibly, limited cognitive ability may constrain a child's performance on a phonological task. For example, Goldberg and Rothermel (1984) and Aram et al. (1984) attempted to administer lexical-decision tasks with very little success. Indeed, Aram et al. abandoned their investigation as the children were unable to grasp the concept of a "pretend" word. A third control group consisting of normally developing children with equivalent reading skills could also be used to investigate the hypothesis that hyperlexic children's phonological skills are a consequence of reading experience.

A natural by-product of adopting a developmental perspective is to ask questions concerning the longer term. For example, a 2-year-old child who can read only a handful of words may well be considered to be hyperlexic, whereas for a 5-year-old child, this level of word recognition would be unremarkable. Although there is considerable evidence to suggest that hyperlexic children have exceptional word recognition, it is not clear whether this is maintained throughout development. Aaron et al. (1990) presented the case of DL, a retarded 19-year-old female. Although she had a developmental history of hyperlexia, her word-recognition development attenuated at the fourth-grade level (see Aram & Healy, 1988, for similar but anecdotal reports). For DL, then, the diagnosis of hyperlexia may no longer be appropriate.

The connectionist perspective outlined above provides a natural framework for thinking about development (Plunkett et al., 1997). In Plaut et al.'s (1996) model, the semantic pathway only had a measurable influence on reading behavior later in development. In hyperlexia, a well-developed phonological pathway may support a reasonable level of word recognition. However, there may come a point at which the development of word recognition may be constrained because of lack of support from the semantic pathway. Longitudinal studies are needed to test the hypothesis that although hyperlexic when tested at a young age, a child's reading skills may reach a peak so that by late childhood, they would no longer be considered hyperlexic.

Summary, Conclusions, and Future Directions

This review has considered the nature of hyperlexic children's reading skills. Generally, hyperlexic children manifest cognitive, language, and social impairments, yet they develop surprisingly good word-recognition skills, considerably in advance of their reading comprehension. It is argued here that their extraordinary word recognition may be seen as a consequence of individual differences in a number of reading-related skills, and as such, their reading development can be well accommodated within a theoretical framework of normal reading development.

Future studies investigating individual differences in reading-related skills will inform our understanding in at least two ways. First, comparisons between hyperlexic children and nonhyperlexic MA-matched control children will reveal to what extent hyperlexic children possess certain strengths and weaknesses, which may explain why they develop surprisingly good word recognition. Various reading-related skills may be relevant: This review has outlined a number of potentially important sources of individual difference such as in phonological processing skills, print exposure, and memory; plausibly, other skills will also be important. Second, to trace the longitudinal relationships between skills in these areas and reading development will provide information concerning the progression of reading development in hyperlexia. Although parental and clinical reports document the very early onset of reading, the nature of these early skills has not been investigated objectively. It may be the case that hyperlexic children's early reading is quite different from that of normally developing children. Likewise, little is known concerning the long-term prognosis of reading skills in hyperlexic individuals.

Finally, this review has focused on the actual reading performance of hyperlexic children; very little research has considered other features of hyperlexia, most notably, the early onset of reading development and the tendency for hyperlexic individuals to become preoccupied with reading. Throughout this review, it has been suggested that these aspects of hyperlexia may be crucial if an individual child is to develop a hyperlexic reading profile. Very clearly, future research needs to examine these factors systematically; accurate and objective measures of these other aspects of hyperlexia may provide clues as to those characteristics that distinguish hyperlexic from nonhyperlexic children.

As is clear from this review, progress toward a greater understanding of hyperlexic children's word-recognition skills is being made. In contrast, our understanding of the nature of reading-comprehension skills in hyperlexia is poor. For most hyperlexic children, their comprehension deficit is consistent with their verbal ability, although for a few children, their reading comprehension may be considerably lower. Notably, most of the studies concerned with reading comprehension have investigated comprehension of isolated words using written versions of the PPVT or BPVS; experimental investigations of sentence-level or text-level comprehension are lacking. Such studies will be important in elucidating how higher-level processes such as the ability to integrate information or draw inferences are related to more basic single-word processing and vocabulary knowledge, both in hyperlexia and in normal reading development. From a practical perspective, a greater understanding of these theoretical issues will contribute to knowledge concerning the clinical diagnosis and educational management of hyperlexic children.

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